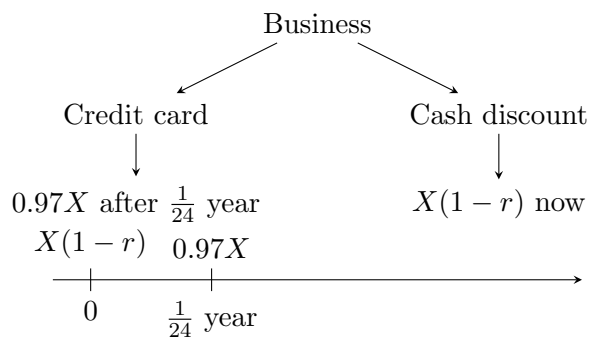


Problem 8.17

A business permits its customers to pay with a credit card or to receive a percentage discount of r for paying cash. For credit card purchases, the business receives 97% of the purchase price one-half month later. At an annual effective rate of discount of 22%, the two payments are equivalent. Find r .

Solution

We can summarize the two payment options with the following diagram:



Let X be the original purchase price. If the customer pays cash, the business receives

$$X(1 - r)$$

immediately. If the customer pays by credit card, the business receives

$$0.97X$$

one-half month later. Since one-half month is $\frac{1}{24}$ of a year, the present value of the credit card payment is

$$0.97X(1 - 0.22)^{1/24}.$$

The two payments are equivalent, so

$$X(1 - r) = 0.97X(1 - 0.22)^{1/24}.$$

Canceling X gives

$$\begin{aligned} 1 - r &= 0.97(1 - 0.22)^{1/24}, \\ r &= 1 - 0.97(1 - 0.22)^{1/24}, \\ r &\approx 0.03997. \end{aligned}$$

Therefore,

$$r \approx 4.0\%.$$